A Cross-Modality Feature Transfer Method for Target Detection in SAR Images

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***Abstract*— Synthetic aperture radar (SAR) ship detection methods have achieved remarkable progress in recent years. However, unlike RGB images, the characteristics of SAR imaging will result in non-intuitive feature representations. Furthermore, due to the insufficient data of SAR images, existing methods relying on plenty of labeled SAR images may be hard to achieve promising performance. To address the aforementioned issues, a cross-modality feature transfer (CMFT) method is proposed in this article, which enhances feature representations in the SAR modality by transferring rich knowledge in the RGB modality. First, we propose a multilevel modality alignment network (MMAN), which encourages the model to effectively learn modality-invariant features and alleviate the large cross-modality discrepancies by aligning features from multilevels (scene level, local level, global level, and instance level). Second, to address the underperformance of samples with non-intuitive features in the modality alignment, we introduce a hard-sample supervision module (HSM) in the stage of feature extraction, which can thoroughly exploit the feature of hard-to-align samples by giving more optimization energy for them. Third, to enhance the discriminability of instance-level features, a feature complementary module (FCM) is customized to fully explore the potential complementary clues between instance-level features and context information for the instance-level feature alignment. Extensive experimental results demonstrate that the CMFT outperforms the state-of-the-art detectors. Compared to the baseline model, CMFT improves the accuracy by 3.1% mean average precision (mAP) on the SSDD dataset and 3.4% mAP on the HRSID dataset, demonstrating its superior SAR ship detection performance.**

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I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging sensor with the capability of observing the Earth all-day and all-weather, and has been extensively applied in the military and civilian fields. As a great branch of SAR image interpretation, SAR ship detection has drawn considerable attention for its pivotal value in ocean monitoring and national defense security in recent years. During the early detection of ships in SAR images, the improved algorithms [1], [2], [3] based on constant false alarm rate (CFAR) are the most widely used methods. But these CFAR-based algorithms rely on human predefined distributions, which are challenging to distinguish targets in the complex scene accurately. Therefore, these methods still suffer from tremendous difficulties in practice.

Starting from the seminal work [4], [5], [6], [7], [8], [9] of deep learning, object detection has gained extensive attention and research focus [10]. There has been significant progress in the ship detection of SAR images [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25]. Most SAR image target detection algorithms aim at improving and innovating the structure and mechanism of feature extractors. These include explorations and researches on feature fusion techniques to obtain more comprehensive and richer semantic information. For example, Wang et al. [11] proposed a multifeature fusion network for the integration of spatial and frequency information to leverage information from different domains. Zhang et al. [12] combined the feature pyramid network with the adaptive spatial feature fusion module to fully utilize the feature information at different scales, thereby enhancing the ability of extracting multiscale features. Xiao et al. [13] proposed an aligned feature fusion module to avoid the deviation of target location when integrating different scale features. Additionally, researchers have focused on incorporating attention mechanisms to assist the model in precisely attending to important objects or key regions. For instance, Bai et al. [14] proposed a united attention module to reduce background noise interference by focusing on the feature information of ship targets from scale and spatial domains. Du et al. [15] proposed a multiscale attention mechanism module, which enhanced spatial and channel attention to enable the network to utilize more valuable information. Apart from improving the feature extractor, some algorithms

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enhance the generalization ability of models by improving the learning strategies. These include adopting multitask learning strategies, optimizing loss functions, and so on. For example, Zhang et al. [16] proposed a ship detector based on multitask learning to learn specific attributes (i.e., the ship texture) through three collaborative tasks. Yang et al. [17] proposed a novel varifocal loss to encourage the network to better distinguish between the objects and the background. Qu et al. [18] proposed a mask loss to make the target positions in the mask feature map more prominent, thereby reducing false alarms. These works borrow the ideas of optical study and have developed relatively mature and diversified. Most of them attach much importance to the design of models and focus on improving the depth, width, and levels of networks. However, compared to the amount of RGB data, the size of SAR datasets is small. A common learning theory tells us that the convergence of empirical risk minimization (ERM) [26] could not be guaranteed if the complexity of the model does not match the size of the training data. In practice, we find it difficult to build a large-scale dataset like ImageNet [27] in the study community of SAR ship detection, resulting in some algorithms generally suffering from limited performance. In addition, due to the specific imaging mechanism and physical limitations of SAR images, single-modality algorithms that solely rely on SAR images for detection have shortcomings such as incomplete information and poor robustness.

Compared with SAR images, the amount of RGB images is enormous, and RGB images have detail-rich information and intuitive feature representations [28], which are easier to observe and interpret. Therefore, how to effectively use the massive RGB images to assist SAR images to learn a knowledge-rich model is a direction deserving of further study. Due to the significant differences in imaging mechanisms and feature representations between the two modalities, it is challenging to achieve cross-modality knowledge transfer. However, many researchers have found that methods based on heterogeneous transfer learning [29] can address the differences in feature space between different modalities and enable feature transformation. The current works that aim at transferring the knowledge from the RGB modality to the SAR modality by heterogeneous transfer can be mainly classified into these three categories.

1) Methods based on fine-tuning pretrained models. These methods learn modality-related knowledge on a large amount of RGB data, and then extract modality-related features to fine-tune the model for SAR modality tasks. For instance, Li et al. [30] utilized a model pretrained on a large-scale RGB dataset PASCAL VOC and fine-tuned it to transfer the learned RGB knowledge to the SAR modality. However, the inherent characteristics (shortening, layover, and shadow) caused by the unique imaging mechanism of SAR images are quite different from the characteristics of RGB images. Therefore, extensive fine-tuning may be needed to achieve good performance when transferring between these modalities. Bao et al. [31] proposed an OSM pretrained technique to learn the common representation of the matched optical and SAR images via the bridge neural network to enhance the universal features embedding of the SAR image. Nevertheless, the matched samples (i.e., pair-wise RGB images and SAR images) for training data are scarce and hard to obtain at present, hence this solution may be suboptimal.

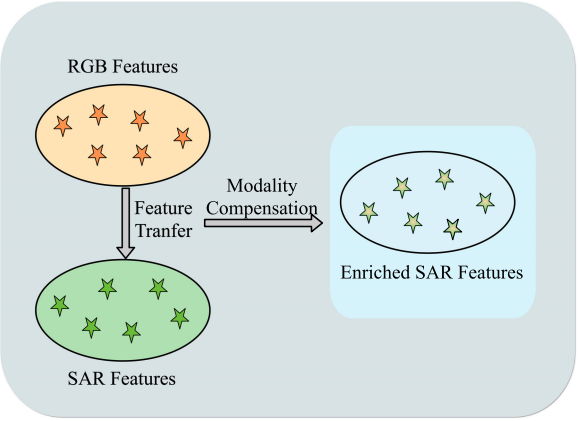


Fig. 1. Our motivation. By transferring the abundant knowledge in the RGB modality, the feature representations of SAR modality is enriched.

2) Methods based on style transfer. One approach is to correspond the features of the RGB modality to the style of the SAR modality, transforming the RGB features into a style that is closer to the SAR modality to reduce the impact of modality discrepancy. By enabling the adaptation of RGB knowledge to the task requirements of the SAR modality, it facilitates the transfer of RGB knowledge. Some studies [32], [33] choose to transfer RGB images into SAR-style images via CycleGAN [34], which alleviates modality discrepancy to some extent. However, these methods only transform RGB images into SAR style, without involving the physical properties of SAR images. These SAR-like images do not exhibit the same scattering characteristics as real SAR images, especially for targets. There exists a large amount of low-quality data in the SAR-like images. In such a case, the results of these methods heavily depend on the quality of the converted SAR-like images and are unstable. And it is more equivalent to a data augmentation of SAR images. Another approach is to convert SAR images to RGB-style images via CycleGAN and then utilize an excellent RGB-based object detection model for detection. However, as some scholars have found in Optical and SAR image registration based on feature decoupling network [35], the SAR images focus more on the scattering characteristics of targets and suffer from severe speckle noise, the conversion of such data to higher quality RGB data cannot achieve the same level of image quality as true RGB data, and many targets cannot be recovered, which is not conducive to subsequent tasks. Therefore, the significant geometric and radiometric differences between the two modalities render the methods based on style transfer become not optimal solutions.

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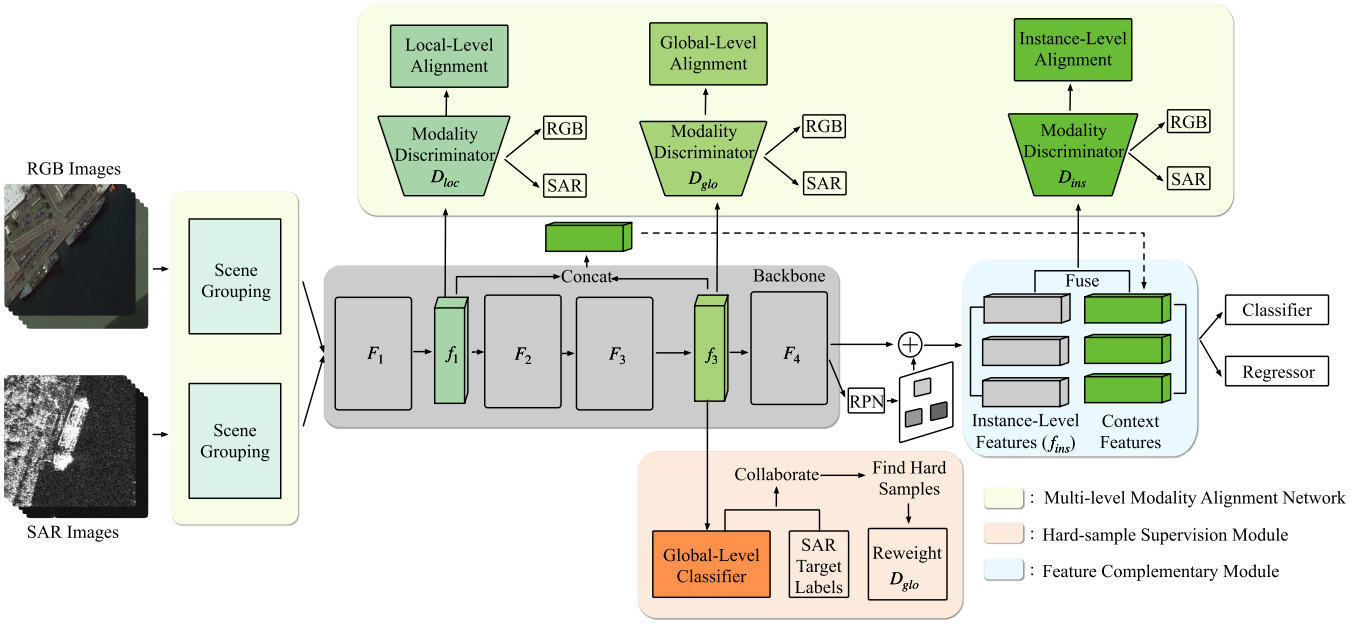


Fig. 2. Overall framework of CMFT, in which the green block diagram is the proposed MMAN, the orange block diagram is the proposed HSM, and the blue block diagram is the proposed FCM.

3) Methods based on feature alignment. Inspired by the domain adaptive algorithm [36], [37], [38], some researchers [39], [40], [41], [42] have devoted their efforts to learning domain invariant representations and aligning the latent feature spaces of the RGB and SAR images. By aligning the features of the RGB modality and SAR modality, this method learns modality-invariant features, making the features of the two modalities closer or more similar in the feature space, thus enabling knowledge transfer. However, there is a widespread latent heterogeneity between the feature spaces of SAR and RGB images, thus the domain adaptation methods may be hard to bridge such large domain gaps. It is undeniable that these detection methods based on domain adaptation provide a feasible insight for us, which achieve the knowledge transmission from RGB images to SAR images in a real sense.

Inspired by the above discussions, a novel framework based on the cross-modality feature transfer (CMFT) is proposed for ship detection in SAR images. The feature representations of SAR images will be enriched by gradually exploring modality-invariant features between RGB images and SAR images (as shown in Fig. 1). CMFT consists of three key components: a multilevel modality alignment network (MMAN), a hard-sample supervision module (HSM), and a feature complementary module (FCM). Different from the previous methods that align the RGB and SAR images without any distinction, MMAN tries to narrow the gap between two modalities from the multilevel and capture their interrelationship in a similar scene. For the hard samples which have underperformance during modality alignment, we present HSM to give more optimization energy for them and guide the backbone network to extract more discriminative features of them. Furthermore, FCM is customized to reinforce the instance-level features for effective alignment by capturing the underlying complementary clues between holistic context information and instance-level features.

In summary, the main contributions of the proposed CMFT are as follows.

1) We propose a MMAN to enrich SAR modality feature representations. By aligning the feature spaces on the scene level, local level, global level, and instance level, the knowledge will be fully transmitted from the RGB modality to the SAR modality. Such a design can effectively reduce the impact of the distribution gap between two modalities and help the SAR modality achieve excellent detection results.
2) We devise a HSM, which combines an additional global-level classifier with labels to give more optimization energy for those hard-to-align samples. By adaptively adjusting their loss weights, samples with non-intuitive features can be focused during modality alignment, and their representations will be enhanced.
3) A FCM is presented to capture the potential complementary clues between instance-level features and context information for the instance-level alignment. FCM promotes the discriminability of the instance characteristics and compensates for their missing features.

II. Methodology

The overall architecture of CMFT is illustrated in Fig. 2, which is improved based on the conventional two-stage detection method Faster R-CNN. First, for input RGB and SAR images, MMAN is employed to group them into the inshore scene and offshore scene to guarantee feature similarity as much as possible for subsequent alignment from the scene level. Once scene grouping is complete, the images will be sent to the backbone network for feature extraction. In the feature extraction stage, we use MMAN to align the outputs

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of the first stage and the third stage of the backbone network at the local level and the global level, respectively. Next, we utilize HSM to process the outputs of the third stage of the backbone network and give more optimization energy for the hard samples. Then, the instance-level features, which are generated by the ROI-pooling layer are processed by FCM to improve their discriminability. After that, the enriched instance-level features will be aligned by MMAN at the instance level. Finally, the information acquired through the aforementioned procedures will be transmitted, respectively, to the anchor regression and classification sub-networks for the task of localization and discrimination. In the following, the proposed modules of CMFT will be elaborated on in detail.

A. Multilevel Modality Alignment Network

The special imaging mechanism of SAR images leads to the problems that its feature representations are not intuitive and the cost of manual labeling is expensive. Therefore, the current amount of SAR image data is an order of magnitude lower in comparison to RGB images. In addition, RGB images contain more detail-rich feature information than SAR images. Thus, we hope to utilize RGB features to compensate for the modality representations of SAR images, so as to establish robust SAR modality features.

However, the difference in imaging mechanism has caused a huge modality gap between the SAR and RGB images, which is one of the main obstacles in the process of knowledge transmission. Since the characteristics of different level features are not the same, the problem of feature transfer caused by modality differences cannot be effectively solved by a single-time operation. To further mitigate this challenging issue, a MMAN is proposed to learn modality-invariant features by mapping the features of two modalities to the same feature space. MMAN progressively transfers knowledge by modality alignment from four levels: scene level, local level, global level, and instance level. As shown in the green block diagram of Fig. 2, the alignment operations work, respectively, at the images, features output from the first stage and the third stage of the backbone network, and features output from ROI-pooling. Through the multilevel alignment operation, the difference between modalities can be reduced from the data to the feature and then to the instance aspect, and the modality-invariant features can be extracted thoroughly.

Specifically, we first group the images at the scene level. Some works [39], [41] align the overall distribution of the RGB and SAR images without any distinction, as shown in Fig. 3(a). However, there are significant differences between cross-modality images, which may bring the transfer of some undesired information, resulting in a negative transfer. We argue that there is more modality-sharing knowledge worth transferring between RGB and SAR images with a similar scene. In other words, modality-invariant features learned from the modality with a similar scene are more helpful to SAR images and contain more relevant information. To reduce the interference caused by image content, we perform a scene grouping on images before feature extraction. In addition, scene grouping also reduces the strict restrictions on data

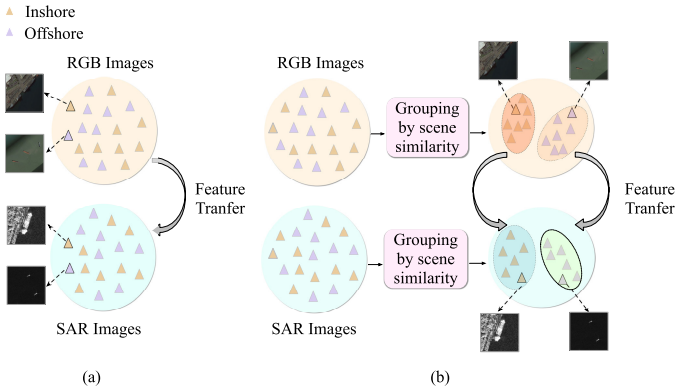


Fig. 3. Schematic of the algorithm for feature transmission without scene grouping and the algorithm for feature transmission after scene grouping. (a) Some algorithms. (b) Our method.

similarity, and greatly enhances the availability of data by improving the similarity of aligned data through the method itself. It could distinguish the inshore or offshore scene of SAR and RGB images by calculating the texture complexity, as shown in Fig. 3(b), which is essentially an alignment at the data level. In Fig. 3(b), the orange triangles represent inshore images, and the purple triangles represent offshore images. The dataset containing a mixture of inshore and offshore images is categorized into two classes after scene grouping, and then the images in each class are aligned separately. The principle of scene grouping is that the scene with higher texture complexity will be divided into the inshore scene, and the scene with lower texture complexity will be divided into the offshore scene. The specific process of scene grouping can be described as follows. To effectively reduce the impact of noise in images, the images are smoothed with Gaussian filtering at first

$$H(x, y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) f(x, y) \tag{1}$$

where $f(x, y)$ represents the initial gray value of pixel (x, y) and $H(x, y)$ represents the gray value of pixel (x, y) after gaussian filtering. Let $H_x(i, j)$ and $H_y(i, j)$ be the gradient values of the two directions of the image after filtering, the gradient value and the gradient direction of the pixel (x, y) are

$$H(i, j) = \sqrt{H_x^2(i, j) + H_y^2(i, j)} \tag{2}$$

$$\theta(i, j) = \arctan\left(\frac{H_y(i, j)}{H_x(i, j)}\right). \tag{3}$$

Then, the gradient amplitude is suppressed by a non-maximum value, and the edge information is screened by the double threshold method. After that, the edge points will be connected and we get the image which is after edge extraction. Computing the complexity of images can facilitate the completion of subsequent image scene grouping tasks. The complexity of the image after edge extraction can be expressed as

$$\lambda = \sum_{x,y} \frac{I(x, y)}{w * h} \tag{4}$$

where $I(x, y)$ denotes the gray value of pixel (x, y) after edge extraction. The w and h indicate the width and height

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of the image. Finally, based on the principle that the more complex the scene, the greater the λ value, the images can be grouped into two scenes: inshore and offshore. The subsequent modality alignment is based on a similar scene, that is, modality alignment in the inshore or offshore scene.

Next, to train a detector capable of extracting modality-invariant features and closing the gap between modalities, we perform alignment between the two modalities at the local, global, and instance levels. The difference in the alignment operation at each level is shown in Fig. 4. The alignment operation is implemented mainly by modality discriminators, which are designed to classify the modality (RGB or SAR) of features, and the feature extractor learns to cheat modality discriminators by extracting modality-invariant features.

As we all know, at the shallow level, features tend to have a relatively smaller receptive field, and the local information on the shallow feature map is abundant. Therefore, for the extracted low-level features (the outputs of the first stage of the backbone network) which are local and general, we perform the alignment at the local level. At the local level, the modality discriminator D_{loc} is used to focus on distinguishing and effectively aligning the features of low level, while our feature extractor in this process will be trained to deceive D_{loc}. In this way, the modality gap between features will be reduced. D_{loc} is a fully convolutional network, and it is worth emphasizing that the kernel size of it is equal to one. And its output of the modality prediction map matches the dimensions of the input feature. We employ a least-squares loss L_{loc} to train the local-level modality discriminator, which can be expressed as

$$L_{\text{loc}_R} = \sum_{i,w,h} D_{\text{loc}}\left(F_1^R\left(x_i^R\right)\right)_{wh}^2 \tag{5}$$

$$L_{\text{loc}_S} = \sum_{i,w,h} \left(1 - D_{\text{loc}}\left(F_1^S\left(x_i^S\right)\right)_{wh}\right)^2 \tag{6}$$

$$L_{\text{loc}} = L_{\text{loc}_R} + L_{\text{loc}_S} \tag{7}$$

where x_i^A denotes the input image, $F_1^A(x_i^A)$ represents the local feature, which is the output of the first stage of the backbone, $D_{\text{loc}}(F_1^A(x_i^A))_{wh}^2$ denotes the output of modality discriminator of local level, w denotes width, h denotes height, and A denotes the modality of data. With the deepening of the network, the receptive field grows larger. At the same time, the deep feature map becomes increasingly enriched with global information. For the extracted high-level features (the outputs of the third stage of the backbone network) which are global and distinct, the alignment is performed at the global level. At the global level, to reduce the difference of deep features with particularity, we train the modality discriminator D_{glo} to align the modality of input features by predicting which modality they come from. The classification loss of D_{glo} is summarized as

$$L_{\text{glo}_R} = -\sum_i \left(1 - D_{\text{glo}}\left(F_3^R\left(x_i^R\right)\right)\right)\log\left(D_{\text{glo}}\left(F_3^R\left(x_i^R\right)\right)\right) \tag{8}$$

$$L_{\text{glo}_S} = -\sum_i D_{\text{glo}}\left(F_3^S\left(x_i^S\right)\right)\log\left(1 - D_{\text{glo}}\left(F_3^S\left(x_i^S\right)\right)\right) \tag{9}$$

$$L_{\text{glo}} = L_{\text{glo}_R} + L_{\text{glo}_S} \tag{10}$$

where $F_3^A(x_i^A)$ denotes the global feature, which is the output of the third stage of the backbone network, and $D_{\text{glo}}(F_3^A(x_i^A))$ denotes the output of the global-level modality discriminator. Since the instance-level features still have intermodality differences, such as appearance, size, perspective, etc. Ignoring these differences will adversely affect the detection results. The instance-level alignment can assist the model in accurately understanding the relationship between the RGB and SAR modalities, enabling it to learn broader rules, and thereby enhancing its generalization ability. Therefore, a modality discriminator D_{ins} is introduced to achieve strict alignment of instance-level features. It is optimized using the loss, given as follows:

$$L_{\text{ins}_R} = -\sum_j \log\left(D_{\text{ins}}\left(f_{\text{ins}}{}_j^R\right)\right) \tag{11}$$

$$L_{\text{ins}_S} = -\sum_j \log\left(1 - D_{\text{ins}}\left(f_{\text{ins}}{}_j^S\right)\right) \tag{12}$$

$$L_{\text{ins}} = L_{\text{ins}_R} + L_{\text{ins}_S} \tag{13}$$

where $f_{\text{ins}}{}_j^A$ represents target proposal feature obtained from ROI-pooling, and $D_{\text{ins}}(f_{\text{ins}}{}_j^A)$ denotes the output of the instance-level modality discriminator.

B. Hard-Sample Supervision Module

The features of SAR are sensitive to parameters such as pitch angles and azimuth angles. When these parameters are changed, it is prone to bring differences in target structure features. Even the contour features of the target are not particularly obvious at some imaging angles. These samples with non-intuitive features may also perform poorly during modality alignment. Some detection algorithms [39], [41], [42] based on feature alignment treat all samples indiscriminately. This is undesirable.

As for hard-to-align samples should be given more optimization energy. Because these samples also play a pivotal role during cross-modality knowledge transmission. Failure to do so can result in the feature information of these samples being overlooked and cannot be fully utilized. Making full and effective use of these hard-to-align samples will further solve the problem of the alignment of modality difficulties existing in cross-modality information transfer. Furthermore, it is worth noting that hard-to-align samples can be particularly beneficial for enhancing the capability of the backbone to process features. In contrast, easy-to-align samples may help the model learn modality alignment knowledge faster, but they may not provide as much unique or informative data as hard-to-align samples and contribute less to the performance improvement of the model. By fully utilizing the feature information from hard-to-align samples, we can improve the detection ability of the network and facilitate a more effective transfer of cross-modality information.

Therefore, we propose a HSM, as shown in the orange block diagram of Fig. 2. Compared with low-level features, high-level features are more representative and have richer semantic information, which typically possess better semantic representation capabilities. By adding a classifier at the third stage of the backbone network to process global-level features which

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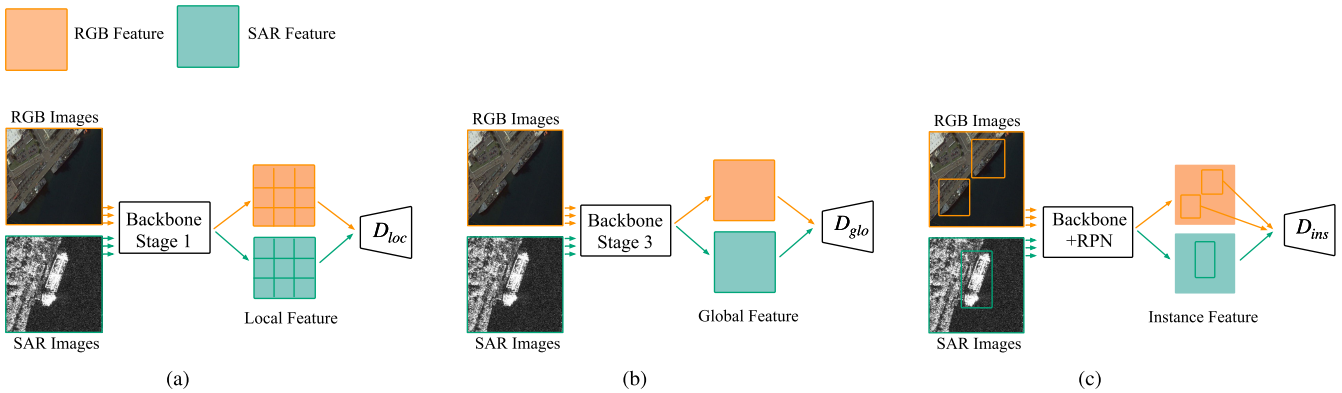


Fig. 4. Comparison of three alignment methods. (a) Local-level alignment. (b) Global-level alignment. (c) Instance-level alignment.

contain more semantic information, the backbone network can be trained to enhance the semantic features of hard-to-align samples at the global level. Then, we combine the classifier with the label information of the target to select hard samples lacking discriminant information. If the classifier predicts that there is no ship in the image, but the label tells us there exists the ship, then the image will be judged as a hard sample. And its classification loss will be adjusted by reweighting during the process of global-level modality alignment. The specific implementation is as follows.

The adaptive average pooling operation is performed on the output layer of the third stage of the backbone, and then the pooled features are classified by a classifier which is implemented by a 1×1 convolution module. In addition, the loss for training this global-level classifier is the binary cross-entropy loss, which is defined as

$$L_c = y_i \log(P_i) + (1 - y_i) \log(1 - P_i) \tag{14}$$

where y_i is the ground truth label and P_i is the predicted. We define the following method to measure the consistency between the labels and global-level predictions:

$$\beta = \begin{cases} e^{1-P_i}, & n \geq 1 \\ e^{P_i}, & n = 0 \end{cases} \tag{15}$$

where n is the number of targets in the image. We use β to reweight the loss of the global-level modality discriminator, so that the hard-to-align samples can get more focus in the modality alignment. The classification loss of D_{glo} is updated as

$$L_{\text{glo}} = L_{\text{glo}_R} + \beta L_{\text{glo}_S}. \tag{16}$$

C. Feature Complementary Module

At present, some methods [39], [40] align the instance-level features which output from ROI-Pooling could alleviate local-instance bias between different modalities to some extent. However, the instance-level features which lack the perception of the holistic context information represent the local targets independently. Therefore, the effective use of the holistic context information can induce instance-level alignment accurately, and it is also crucial for subsequent detection tasks. Furthermore, instance-level features obtained from ROI-pooling are specific and differ, and the contextual information obtained before ROI-pooling is relatively invariant across modalities. Thus, the instance-level features and the holistic context information could be complementary. A sufficient fusion of them could make a more accurate and comprehensive representation of the object of interest, thereby more accurately guiding instance-level alignment. However, how to effectively integrate the two remains a technological problem that needs to be addressed.

Some algorithms [36], [38] have also realized this issue and made improvements, but the majority of them choose to fuse the contextual features with the instance-level features by concatenation. Although these methods can enhance the instance-level alignment effect to a certain extent, these two features are independent of each other, and the underlying complementary relationship between them will be ignored, which is vital for modality alignment. Furthermore, since the instance-level features obtained from the two modalities are asymmetric, the common fusion methods may cause the gradient explosion. To address these issues, we customize a FCM (as shown in the blue block diagram of Fig. 2) that combines contextual features with instance-level features reasonably for aligning the instance-level feature representations effectively. First, we denote context vector outputs from the first stage and the third stage of the backbone as f_1 and f_3, and concatenate the two vectors as a new vector f_4 to obtain the holistic context information. Then, to enable information interaction, we leverage a nonlinear fusion strategy. The formulation is as follows:

$$f_{\text{fus}} = \frac{1}{\sqrt{c}}(b_1 f_4) \odot (b_2 f_{\text{ins}}) \tag{17}$$

where $\odot$ represents the Hadamard product, c denotes the dimension of f_{fus}. The b_1 and b_2 represent random matrices, and each of their elements follows a symmetric distribution with univariance. We leverage the randomized methods [43], [44] to estimate the tensor product unbiasedly. Finally, the instance-level alignment loss is updated as

$$L_{\text{ins}_R} = -\sum_j \log\left(D_{\text{ins}}\left(f_{\text{fus}}{}_j^R\right)\right) \tag{18}$$

$$L_{\text{ins}_S} = -\sum_j \log\left(1 - D_{\text{ins}}\left(f_{\text{fus}}{}_j^S\right)\right) \tag{19}$$

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$$L_{\text{ins}} = L_{\text{ins}_R} + L_{\text{ins}_S}. \tag{20}$$

The detection task loss of our method can be written as L^A_{det}. So, the overall loss of CMFT is expressed as follows:

$$L = L^R_{\text{det}} + L^S_{\text{det}} + \partial\left(L_{\text{loc}} + L_{\text{glo}} + L_{\text{ins}} + L_c\right) \tag{21}$$

where ∂ represents a weighting factor, it is used to balance the loss of detection task and discriminators. The default value for ∂ is 0.1.

III. Experiments

In this section, we will introduce the dataset, experimental setup, and evaluation metrics used in the experiment. To verify the superiority of our design, we evaluated our model on two different datasets and compared the performance with SOTA and some prevalent models. We also visualize the detection results, feature maps, and t-SNE [45] results of the feature distribution. Additionally, we analyze the parameter of our detector to optimize it for better performance. Furthermore, ablation experiments are conducted by removing some added modules from our detector to assess their influence on the overall performance. A detailed explanation will be provided in the following.

A. Experimental Data Description

The design of our experiment simultaneously trains and aligns the data of RGB and SAR, and tests on SAR images. Our method leverages fully shared networks for RGB and SAR modalities. The RGB ship detection dataset is the HRSC2016 [46], and the SAR ship detection datasets are the most widely used dataset SSDD [47] and HRSID [48].

1) *HRSC2016:* HRSC2016 dataset consists of 1061 RGB aerial images. The sizes of images in the HRSC2016 dataset range from 300 × 300 to 1500 × 900. In our experiment, the HRSC2016 is trained as the RGB modality.
2) *SSDD:* In SSDD, there are 1160 images with approximately 500 × 500 image size, acquired from Sentinel-1, TerraSAR-X, and RadarSat-2. SSDD contains 2540 ships. We divide the number of training sets and test sets according to 8:2. And the image is designated as the test set if the last number of its image name is 1 or 9, and the remaining images are used for training. In our experiment, the SSDD is trained and tested as the SAR modality.
3) *HRSID:* The HRSID dataset consists of 5604 images with the resolution of 0.5, 1, and 3 m. It contains 16 951 ships and each image has 800 × 800 pixels. For the HRSID dataset, 65% of the images are used for training, and 35% are used for testing. In our experiment, the HRSID is trained and tested as the SAR modality.

B. Evaluation Metrics

To evaluate the detection performance of various detection methods, we utilized three evaluation indices, i.e., the precision (p), recall (r), and mean average precision (mAP)

$$p = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{22}$$

$$r = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{23}$$

where TP, FP, and FN, respectively, represent the number of true positives, false positives, and positive negatives. The mAP is a widely adopted performance metric for evaluating the efficacy of object detection models. It is a comprehensive measure that accounts for both precision and recall, and provides a global evaluation of the ability of the model to accurately detect objects. As such, mAP is frequently used as the primary evaluation criterion in the field of object detection

$$\text{mAP} = \int_0^1 p(r)dr. \tag{24}$$

The $p(r)$ stands for the precision–recall curve.

C. Implementation Details

All experiments were conducted using a consistent hardware platform, which included a GPU (GTX-3090), CPU (Intel 4210R), and 32G of memory. The experimental environment of CMFT is PyTorch 1.10.0, CUDA 11.1 and cuDNN 11.1, and Python 3.7. Our implementation of CMFT was established on the Faster R-CNN [4]. Furthermore, Swin transformer [49] is selected as the backbone, and the FPN [5] structure is used as the neck. We set the ∂ (in overall loss function) and the IoU threshold as 0.1 and 0.5, respectively. To ensure the fairness of comparisons, all ship detectors, ours included, are implemented under the MMDetection [50] toolbox.

D. Performance Comparison

The current part assesses the efficacy of our method based on the CMFT. Table I shows the performance comparison of some state-of-the-art methods on SSDD quantitatively. It demonstrates that CMFT achieves superior performance with the best mAP of 97.3%. The Mor-FP Yolo v4-tiny [51] detector is ranked as the second-best performing detector among other publicly available results at this stage. However, the mAP of CMFT is 0.9% higher than which evaluated on the Mor-FP Yolo v4-tiny. In the meantime, it can be observed that the precision, recall, and mAP of CMFT are 5.4%, 7.7%, and 7.6% higher than those evaluated on the Faster R-CNN, which proves the effectiveness of our improvement. It is worth noting that methods Faster R-CNN [4], PANET [52], Cascade R-CNN [53], Double-Head R-CNN [54], Grid R-CNN [55], DCN [56], ARPN [21], Quad-FPN [22], ASAFE [57], YOLOv4-LITE-MR [58], MDSD [20], and Mor-FP Yolo v4-tiny exclusively employ SAR images from a single modality as their training sets. The detection performance of these single-modality detection algorithms is limited by the inherent characteristics of SAR images, which may be the potential factor contributing to their poor performance.

Consequently, we proceed to analyze and compare the detection performance of four methods aimed at using both RGB and SAR knowledge for SAR image detection. Methods PM Faster R-CNN, SA Faster R-CNN, CG Faster R-CNN, and MA Faster R-CNN are reproduced based on the most used strategies (RGB pretrained model [31], style transfer-based

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TABLE I
EVALUATION INDICES OBTAINED BY DIFFERENT ALGORITHMS ON SSDD

No.	Method	p	r	mAP
1	Faster R-CNN	0.870	0.904	0.897
2	PANET	0.868	0.919	0.911
3	Cascade R-CNN	0.941	0.908	0.905
4	Double-Head R-CNN	0.869	0.919	0.911
5	Grid R-CNN	0.877	0.897	0.889
6	DCN	0.862	0.930	0.922
7	ARPN	0.854	0.906	0.898
8	Quad-FPN	0.895	0.957	0.952
9	ASAFE	0.885	0.959	0.951
10	YOLOv4-LITE-MR	0.940	0.925	0.950
11	MDSD	0.948	0.961	0.963
12	Mor-FP Yolo v4-tiny	0.935	0.886	0.964
13	PM Faster R-CNN	0.906	0.951	0.942
14	SA Faster R-CNN	0.839	0.957	0.943
15	CG Faster R-CNN	0.847	0.959	0.947
16	MA Faster R-CNN	0.866	0.959	0.949
17	Ours	0.924	0.981	0.973

sample supplement method [59], style transfer-based alignment method [33], and modality alignment-based method [39]) to transfer RGB knowledge to the SAR modality. To ensure fairness, we use the same experimental configuration, backbone network, and pretrained model as our method.

Specifically, the PM Faster R-CNN only uses a pretrained model which is trained on RGB images, while the training data is still only SAR images. This method leverages the advantages of a pretrained model on RGB images and enhances its performance in the SAR image modality through fine-tuning, which is equivalent to our baseline model.

The SA Faster R-CNN converts RGB images into SAR-like images through the CycleGAN and then uses them as supplementary samples of SAR annotations. This method realizes the data-level supplement of scarce SAR images by transferring a large number of RGB images into SAR image style.

The CG Faster R-CNN converts RGB images into SAR-like images through the CycleGAN and then aligns the features of the SAR-like and SAR images. This method achieves the transformation between different modalities by performing style transfer to ensure their similarity and then conducts feature alignment.

The MA Faster R-CNN directly aligns the features of RGB and SAR images without any additional operations. Indeed, it is relatively simple to implement and helps to preserve the original characteristics of each modality while maintaining the consistency and correspondence between them.

The commonality between the four cross-modality detectors and CMFT is the knowledge of the RGB modality has been used in the training process. But there exist overlooked deficiencies in these feature transfer-based methods and the lack of further optimization. The PM Faster R-CNN requires extensive fine-tuning to achieve good detection performance. As shown in Table I, the mAP of PM Faster R-CNN is only 94.2%. SAR-like images have some differences in detail and features compared to real SAR images, which may not provide enough information and introduce biases for subsequent detection tasks, thereby affecting the detection results. Thus, the SA Faster R-CNN, which directly uses SAR-like images as auxiliary samples achieves an accuracy of 94.3% mAP. Basically, the detection performance is not improved. For CG Faster R-CNN, although it improves by 0.4% compared with SA Faster R-CNN, it only obtains an accuracy of 94.7% mAP, which is still 2.6% lower than our method. The solution provided by the MA Faster R-CNN for handling the significant differences between the RGB and SAR modalities is not sufficiently comprehensive, which can potentially impact the transmission effectiveness. It is still far lower than our detection results and gets an accuracy of 94.9% mAP. But it achieves the best detection accuracy among these four methods, which proves that modality alignment of RGB and SAR images is the most effective. We also use the CycleGAN model to convert SAR images into RGB-like images and try to use the powerful RGB-based model to achieve object detection. The detection results (less than 30% mAP) are not ideal, even worse than the detection of SAR images directly using the RGB-based model. Furthermore, in our experimental settings and hardware platform, the inference speed of CMFT is 11.9 frames/s and its model complexity is 111.4 GFLOPs. Overall, CMFT achieves excellent detection accuracy among all methods. This phenomenon demonstrates that the features of RGB modality have been successfully migrated to SAR modality, and the detection ability of the network has been enhanced.

To demonstrate the superiority of our proposed algorithm, we also compare CMFT with SOTA and some prevalent approaches on HRSID. Table II shows the experimental results on HRSID quantitatively. As expected, the effect of CMFT on the HRSID dataset is still superior. CMFT achieves the highest 89.6% mAP. The second-best detector is MTL-Det [16], but MTL-Det is still 0.4% lower mAP than CMFT. The third-best detector is improved-FCOS [17], but improved-FCOS is still 1.0% lower mAP than CMFT. In the meantime, by comparing the results with the other four detectors that also utilize RGB knowledge, it is evident that our

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TABLE II
EVALUATION INDICES OBTAINED BY DIFFERENT ALGORITHMS ON HRSID

No.	Method	p	r	mAP
1	Faster R-CNN	0.814	0.819	0.806
2	PANET	0.812	0.829	0.816
3	Cascade R-CNN	0.871	0.822	0.812
4	Double-Head R-CNN	0.817	0.833	0.820
5	Grid R-CNN	0.904	0.809	0.794
6	DCN	0.814	0.834	0.820
7	ARPN	0.857	0.838	0.817
8	Quad-FPN	0.879	0.872	0.861
9	ASAFE	0.834	0.867	0.851
10	Improved-FCOS	0.817	0.901	0.886
11	MTL-Det	-	-	0.892
12	PM Faster R-CNN	0.756	0.882	0.862
13	SA Faster R-CNN	0.742	0.884	0.866
14	CG Faster R-CNN	0.745	0.888	0.869
15	MA Faster R-CNN	0.746	0.893	0.873
16	Ours	0.813	0.911	0.896

TABLE III
COMPARISON OF THE RESULTS ACHIEVED UNDER VARIOUS CONFIGURATIONS OF THE WEIGHT COEFFICIENT ∂ WITHIN THE OVERALL LOSS FUNCTION

Weighting factor ∂	p	r	mAP
0.1	0.924	0.981	0.973
0.5	0.927	0.964	0.958
1.0	0.907	0.968	0.960
1.5	0.923	0.970	0.959
2.0	0.879	0.968	0.955

method, based on CMFT, outperforms them by a large margin, even across different datasets. Besides, under the condition that the precision is basically the same, compared to the faster R-CNN, CMFT improves the recall and mAP by 9.2% and 9.0% points. Our proposed CMFT is always in the first place, verifying the proposed network has a strong capability to investigate the features in SAR images and has excellent robustness. It is worth noting that the similarity of our datasets is insufficient compared with the matched SAR-RGB data. But CMFT still achieved such excellent performance, which is mainly because our method can fully capture transferable information from multilevel latent feature spaces by multilevel alignment. In addition, the excellent performance is also own to the scene grouping can improve the similarity of the aligned data.

Fig. 5 exhibits the qualitative results of SSDD. In Fig. 5, CMFT is compared with Faster R-CNN, double-head R-CNN, DCN, CG faster R-CNN, and MA faster R-CNN, due to limited pages. Ground truths of targets are indicated by green boxes, while the detection results are indicated by red boxes. Fig. 5(a) represents the results of the faster R-CNN. Fig. 5(b) represents the results of double-head R-CNN. Fig. 5(c) represents the results of DCN. Fig. 5(d) represents the results of CG Faster R-CNN. Fig. 5(e) represents the results of MA faster R-CNN. Fig. 5(f) represents the results of CMFT. From the detection results of the first three detectors, it can be observed that the three detectors are not ideal and are prone to be disturbed by the background of land targets and sea reefs to cause a considerable number of false alarms. In contrast, the latter three detectors, which transfer RGB knowledge to SAR modality in the training process, can effectively avoid false alarms. Notably, compared to CMFT, the other two detectors (transfer RGB knowledge to SAR modality in the training process) are not sensitive to detecting some small-scale targets. The detection results suggest that there exist plenty of missed detections. For the pain points that other algorithms fail to solve, the CMFT has been effectively solved. Our proposed CMFT algorithm outperforms all comparison methods. Most significantly, even for targets located at the boundary between the sea and the land, CMFT achieves promising performance with relatively few false alarms. Moreover, under the condition of satisfying the minimum false alarm rate, our algorithm CMFT can not only effectively detect targets of normal scale but also accurately identify targets of small scale. It implies that our algorithm has robust anti-noise performance and keen judgment merits.

E. Parameter Analysis

In this article, our algorithm involves one important hyperparameter: the weighting factor ∂. Its main function is to balance the loss of detection and the loss of modality discriminators. To assess the superiority of the hyperparameter setting in our proposed algorithm, we set several values of the parameter ∂ for analysis and comparison, as shown in Table III, which are 0.1, 0.5, 1.0, 1.5, and 2.0. By examining the performance metrics at various values of ∂, it can be deduced that as ∂ increases, the overall detection efficacy initially decreases, then undergoes a slight rebound, before continuing to decline. Especially when the value of ∂ is set to 2, there is an obvious performance drop, the precision, recall, and mAP are 4.5%, 1.3%, and 1.8% lower than those evaluated when the value of ∂ is set to 0.1. This is mainly because the optimization direction of the network is determined by

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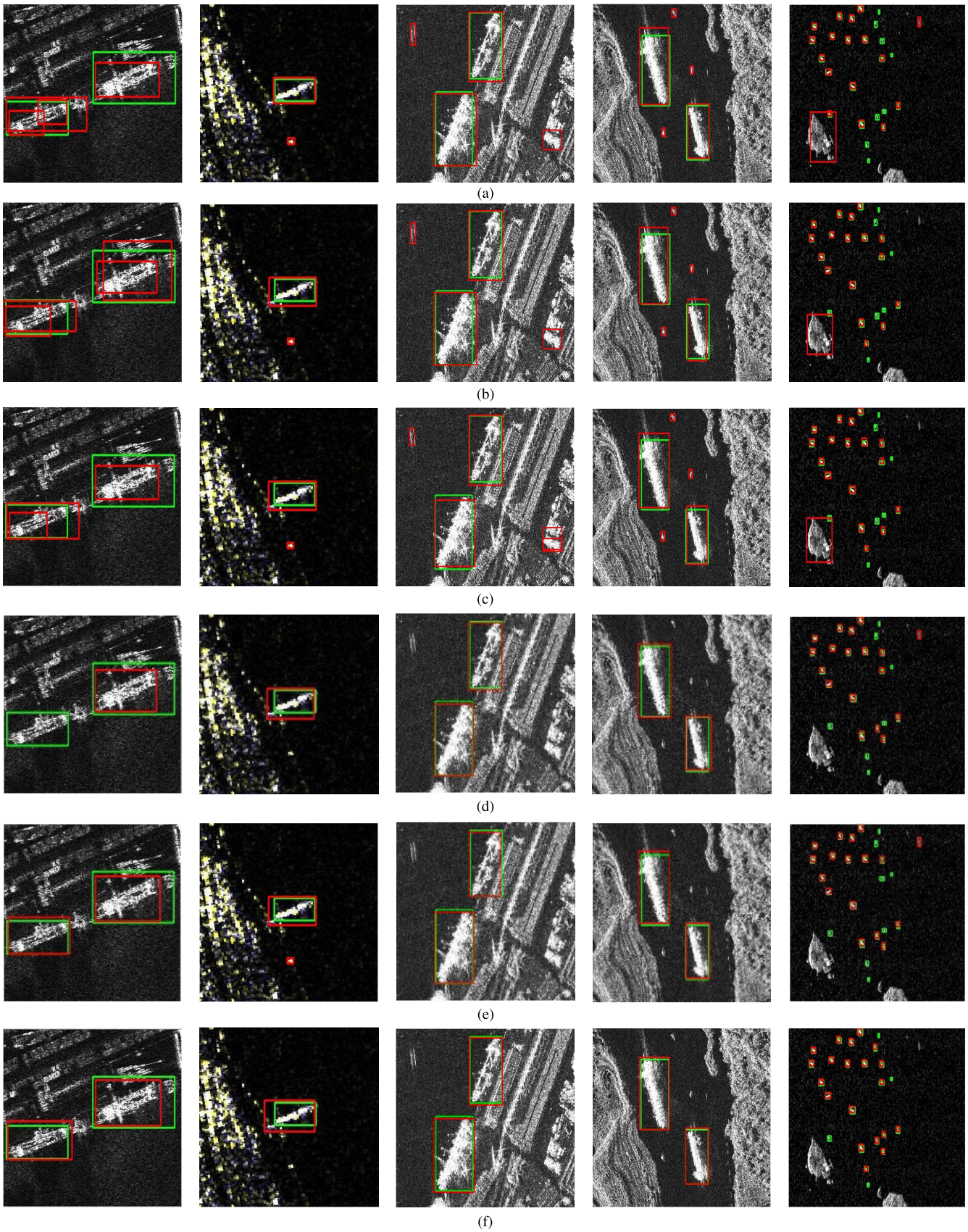


Fig. 5. Visual detection results of CMFT and some comparison methods on SSDD. The green boxes indicate the ground truths of targets, while the red boxes indicate the detection results. (a) Faster R-CNN. (b) Double-Head R-CNN. (c) DCN. (d) CG Faster R-CNN. (e) MA Faster R-CNN. (f) CMFT.

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TABLE IV
ANALYSIS OF CMFT IN ABLATION STUDIES

MMAN	HSM	FCM	p	r	mAP
✗	✗	✗	0.906	0.951	0.942
✓	✗	✗	0.912	0.961	0.952
✓	✓	✗	0.921	0.968	0.958
✓	✗	✓	0.923	0.970	0.961
✓	✓	✓	0.924	0.981	0.973

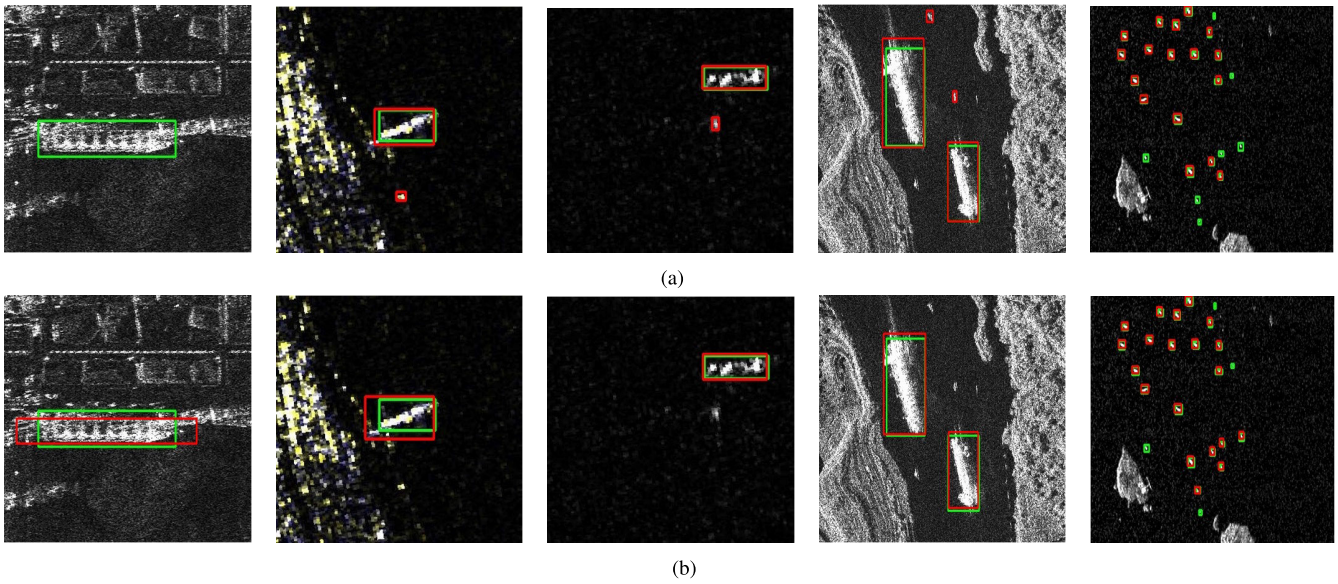


Fig. 6. Visual detection results of (a) baseline model and (b) CMFT on SSDD.

the loss function. And the ultimate goal of our algorithm is the object detection task. Namely, during the training process, emphasis should be placed on detection loss. The goal of modality discriminator loss is to assist the detection task. In the end, in our CMFT algorithm, we establish the value of ∂ as 0.1, which is the optimal value for achieving the best overall detection performance.

F. Ablation Study

In this section, we present a series of ablation studies on CMFT to investigate the effectiveness of each individual component, i.e., MMAN, HSM, and FCM. It is worth emphasizing that Swin transformer is selected as the backbone network and FPN is used as the neck for all the networks in the ablation study. To ensure that the experimental results are fair and accurate, we maintain consistent hyperparameters throughout the validation process. In addition, to increase the effectiveness of the pretrained model, we choose to use the HRSC2016 pretrained model, rather than those obtained from ImageNet or COCO. We argue that extraordinary performance on an excellent baseline model is more reflective of the superiority of our algorithm. In Table IV, "✓" indicates that a certain module is added, while "✗" indicates that the module is not added.

First, as shown in Table IV, the overall rationality of CMFT will be discussed. In the process of module accumulation, the results of our various evaluation indicators have been significantly improved. This finding conveys that each module of our design is compatible, and the overall experimental effect will not be affected by the stacking of modules. Second, we conduct an ablation of MMAN. Surprisingly, the addition of MMAN can achieve 95.2% mAP, which is significantly higher than the baseline model. Its precision, recall, and mAP are 0.6%, 1.0%, and 1.0% higher than those evaluated on the baseline. It has been verified that MMAN is actually what we need to guarantee reasonable performance. Supporting the claim that the feature transfer from the RGB modality to the SAR modality through the MMAN is helpful in delivering highly competitive performance. Since the MMAN module we proposed runs through the entire network, the other two modules in our framework primarily focus on optimizing the modality alignment operation. Therefore, our subsequent analysis of the other two modules is based on the existence of MMAN. Third, we analyze the effectiveness of HSM. It is evident that the network continues to achieve satisfactory results after the addition of HSM, which improves precision, recall, and mAP by 0.9%, 0.7%, and 0.6%. When HSM is removed, the detection result has a significant discrepancy than it is used. In addition, when the FCM module exists, HSM can effectively detect more correct targets. This is mainly due to the fact that giving more optimization energy for hard-to-align samples during modality alignment is available. Finally, by adopting the FCM module to fuse contextual features and

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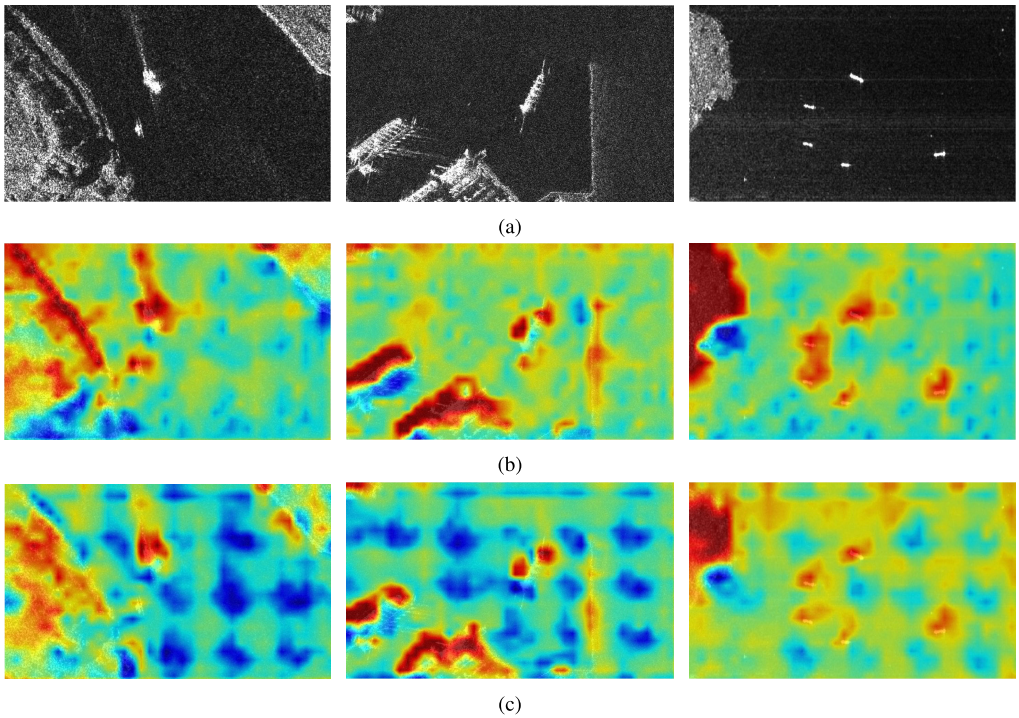


Fig. 7. Visual feature maps of CMFT and the baseline model. (a) SAR images. (b) Baseline model. (c) CMFT.

instance-level features, our algorithm can achieve remarkable detection accuracy. The precision, recall, and mAP are 1.8%, 3.0%, and 3.1% higher than those evaluated on our baseline model. This phenomenon demonstrates that FCM is valuable. By conducting ablation research, the feasibility of CMFT has been substantiated, thus signifying the success of knowledge transfer from RGB to SAR images.

To more intuitively reflect the performance improvements of our algorithm, we have presented visual comparison results between the baseline model and our proposed CMFT, as shown in Fig. 6. The green boxes indicate the ground truths of targets, while the red boxes indicate the detection results. Fig. 6(a) displays the detection results obtained by the baseline model, while the detection results achieved by CMFT are shown in Fig. 6(b). As expected, we can observe that the results generated by the CMFT are more accurate than the baseline model. Regardless of whether the target is large or small in scale, CMFT can detect most of them, while they are ignored by the baseline model. This confirms that our CMFT algorithm effectively transfers sufficient knowledge from RGB images to SAR images, leading to more promising detection results than the baseline. It can also be noticed that, even in some complex backgrounds, CMFT always shows better performance. This may be attributed to the FCM module which makes the instance-level targets more discriminative so that the network can accurately locate the targets. Overall, these visual comparison results provide strong evidence that target detection accuracy in SAR images is significantly improved by our proposed CMFT algorithm, especially in challenging scenarios with complex backgrounds.

As mentioned earlier, CMFT explores modality-invariant features between the RGB and SAR modalities, enabling to find of more suitable feature representations for the SAR modality by efficiently transmitting RGB features. In Fig. 7, we visualize their intermediate feature maps. Fig. 7(a) represents the SAR images, while the feature maps generated by the baseline model are shown in Fig. 7(b), and the feature maps generated by the CMFT algorithm are shown in Fig. 7(c). It is evident that after CMFT processing, the SAR features are more prominent and the background noise is effectively suppressed. These results strongly suggest that our CMFT algorithm significantly enhances the feature representation capability of SAR images while mitigating the effects of background noise, which ultimately leads to more accurate performance.

To further validate the efficacy of CMFT in terms of modality alignment, we provide the feature distribution plot of the two modalities (RGB and SAR) via t-SNE. The features output from the neck. Since the targets in our datasets are single-class targets, our all features are mainly concentrated in one cluster, rather than having multiple clusters with large interclass distances like multiclass targets. In these plots, each red dot corresponds to an image feature in the RGB dataset, while each green dot represents an image feature in the SAR ship dataset. Fig. 8(a) shows the result of mapping the features without modality alignment to a 2-D space. Fig. 8(b) shows the result after modality alignment. From Fig. 8(a), it can be observed that the distribution of features in the 2-D space between the SAR and RGB images is entirely dissimilar and exists in a rather scattered state. This confirms that there is a significant distribution gap between the SAR and RGB modalities due to the distinct dissimilarities between the SAR and RGB images. However, after applying our method, the gap between the two modalities is obviously shortened and the

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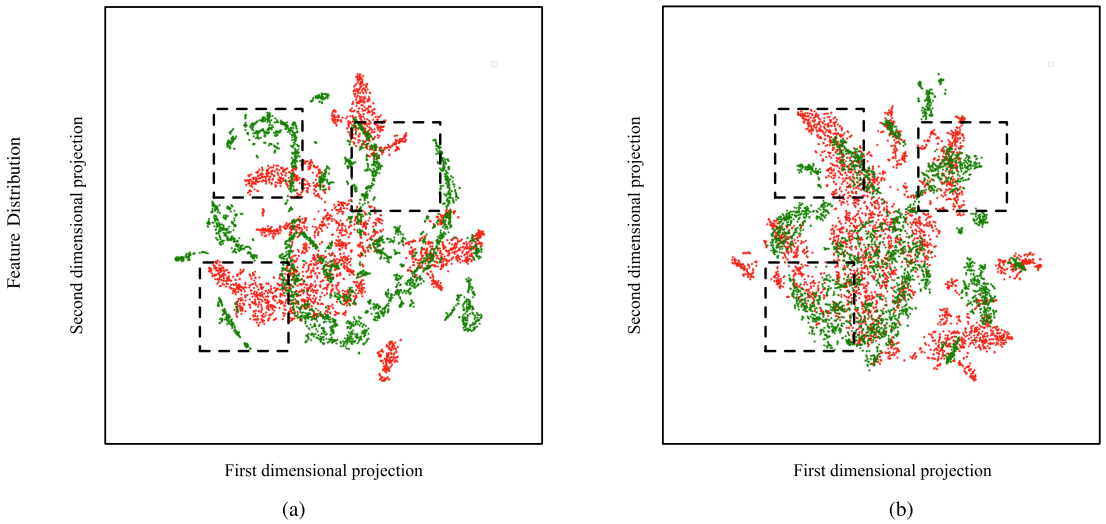


Fig. 8. t-SNE results of the feature (output from the neck) distributions are presented in a visualization, where each red dot represents an image feature from the RGB dataset and each green dot represents an image feature from the SAR ship dataset. (a) Baseline model. (b) CMFT.

features from both modalities form tight clusters, indicating their feature distributions become similar. Therefore, it shows that our method achieves high-quality modality alignment and superior performance, which powerfully proves the predominance of our algorithm.

IV. Conclusion

In this article, we propose a novel CMFT method for ship detection in SAR images, which can transfer abundant knowledge from the RGB to the SAR modality. First of all, to enrich feature representations in SAR images, we customize a MMAN, aiming to exploit modality-invariant representations between RGB and SAR images. Second, we propose a HSM to improve the performance of hard-to-align samples during the modality alignment stage by giving more optimization energy for them. Third, a FCM is designed to leverage the potentially complementary properties between context features and instance-level features, which reasonably fuse these two features to enhance the discriminability of instance-level features for the instance-level modality alignment. In the process of module accumulation, the three modules gradually increase the mAP by 1.0%, 0.6%, and 1.5%. The other various evaluation indicators have also been significantly improved, which directly proves the rationality of our design. Overall, abundant quantitative experiment results demonstrate that CMFT surpasses the classic and SOTA algorithms by a large margin. The qualitative comparisons between different methods suggest that CMFT achieves more visually pleasant detection results.

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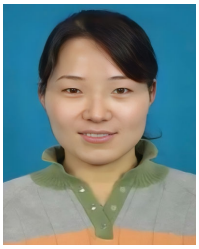
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